

Perfect Competition

1 Perfect competition

1. Definition: **A perfectly competitive market** consists of firms that produce identical products that sell at the same price. Each firm's volume of output is so small in comparison to the overall market demand that no single firm has an impact on the market price. Four characteristics:

- (a) Undifferentiated products;
- (b) Perfect information;
- (c) Large number of buyers and sellers, each buyer or seller has imperceptible effect on market price;
- (d) Free entry and exit.

2. Market demand: The market, or aggregate, demand function is the horizontal sum of the individual (or segment) demands. In other words, market demand is obtained by adding the quantities demanded by the individuals (or segments) at each price and plotting this total quantity for all possible prices.

Let $\mathcal{I}$ index the set of individual buyers and $q^i(P, P_-, y^i)$ be i 's demand of good q as function of its own price P and prices for other goods P_- . Market demand

$$q^d \equiv \sum_{i \in \mathcal{I}} q^i(P, P_-, y^i).$$

3. Property of market demand: homogeneous of degree zero in all prices and the vector of buyers' income.

1.1 Short-run equilibrium

1.1.1 Supply function

1. Short-run profit-maximization:

- (a) If the firm chooses to produce a positive output, $P = SMC$ defines the short run supply curve of the firm. But the firm will choose to produce positive output if and only if $P > AVC$.
- (b) The price below which the firm would opt to produce zero is called the shut down price, P_s . In this case, P_s is the minimum point on the AVC curve.

2. Example: Deriving short-run supply curve

$$SC = 100 + 20q + q^2.$$

$$AVC = 20 + q, \quad SMC = 20 + 2q.$$

Hence $P_s = 20$. The short-run supply curve

$$q = \begin{cases} 0 & \text{if } P < 20. \\ -10 + \frac{1}{2}P & \text{if } P \geq 20. \end{cases}$$

3. The *short-run market supply* at any price is the sum of quantities supplied by all firms in the market. The short-run market supply curve is the horizontal sum of the individual firm supply curves.

Let $\mathcal{J}$ index the set of firms in the market, and $q^j(P, W)$ be the supply function of firm j ,

$$q^s \equiv \sum_{j \in \mathcal{J}} q^j(P, W).$$

1.1.2 Short-run equilibrium

- 1. Definition: A competitive market is in short-run equilibrium at price P^* when $q^d(P^*) = q^s(P^*)$.
- 2. Example: Market demand $q^d(P) = 60 - P$. 300 firms in the market. Each firm has a short-run cost function $SC = 0.1 + 150q^2$.

$$SMC = 300q, \quad AVC = 150q.$$

$$q^j(P) = \frac{P}{300}, \quad q^s = \sum_{j=1}^{300} q^j = P.$$

Market equilibrium

$$q^s = P = q^d = 60 - P \implies P = 30.$$

- (a) Comparative statics: increase in number of firms.
- (b) Comparative statics: increase in demand.

1.2 Long-run equilibrium

1. In the long run, firms can enter or exit the market, which implies a zero profit condition $\pi^j(P^*) = 0$.
2. In equilibrium, market demand equals supply at price P^* . Hence the long-run equilibrium conditions are

$$q^s(P^*) = q^d(P^*) \quad \text{and} \quad \pi^j(P^*) = 0.$$

3. Example: Long-run equilibrium, for each firm

$$AC(q) = 40 - q + 0.01q^2, \quad MC(q) = 40 - 2q + 0.03q^2.$$

Market demand $q^d(P) = 25,000 - 1,000P$.

Each firm j 's long run supply curve

$$40 - 2q^j + 0.03(q^j)^2 = P,$$

Zero economic profit

$$40 - q^j + 0.01(q^j)^2 = P^*.$$

Supply equals demand

$$nq^j(P^*) = 25,000 - 1,000P^*.$$

Solving the equation system gives

$$q^j = 50, \quad P^* = 15, \quad n^* = 200.$$

2 Welfare economics

1. We have seen how to calculate the competitive equilibrium: the price at which supply equals demand. In this section, we investigate the welfare properties of this equilibrium.
2. We first use the representative consumer approach. Suppose that the market demand curve, $x(p)$, is generated by maximize the utility of a single representative consumer who has a utility function of the form $u(x) + y$. The x -good is the good under examination in this particular market. The y -good is a proxy for "everything-else." The most convenient way to think of the y -good is as money left over for purchasing other goods after the consumer makes the optimal expenditure on the x -good.

3. Solving the consumer's problem, we obtain the demand function $p = u'(x(p))$. Note in the case of quasilinear utility the demand function is independent of income. This feature makes for especially simple equilibrium and welfare analysis.
4. As long as we have assumed the representative consumer, we may as well assume a representative firm, and let it have the convex cost function $c(x)$ with $c(0) = 0$. The profit-maximizing supply of the firm is given by $p = c'(x)$. Hence, the equilibrium level of output of the x -good is simply the solution to the equation $u'(x) = c'(x)$.

2.1 Welfare analysis

1. Suppose instead of using the market mechanism to determine the output, we simply determine directly the amount of output that maximized the representative consumer's utility. This problem can be stated as

$$\begin{aligned} \max_x \quad & u(x) + y \\ \text{s.t.} \quad & c(x) + y = \omega. \end{aligned}$$

where ω is the consumer's initial endowment.

2. Substituting from the constraint, we rewrite the problem as

$$\max_x u(x) + \omega - c(x)$$

the first-order condition is $u'(x) = c'(x)$.

3. Therefore, the competitive market result in exactly the same level of production and consumption as does maximizing utility directly. The welfare maximizing problem is simply to maximize total utility. Note that ω is a constant, so our social objective function could simply be $u(x) - c(x)$. Choosing x to maximize $u(x) - c(x)$ is equivalent to choosing x to maximize the area under the demand curve and above the supply curve.
4. Here is another way to look at the same calculation. Let $CS(x) = u(x) - px$ be the consumer's surplus. Similarly, let $PS(x) = px - c(x)$ be profit, or the producer's surplus. Then the maximization of total surplus entails

$$\max_x CS(x) + PS(x) = [u(x) - px] + [px - c(x)] = u(x) - c(x).$$

Hence, we can also say that the competitive equilibrium level of output maximizes total surplus.

2.2 Pareto efficiency

1. We have just seen that a competitive equilibrium maximizes the sum of utilities, at least in the case of quasilinear utilities. A more general objective is the idea of Pareto efficiency.
2. A Pareto efficient allocation is one for which there is no way to make all agents better off. Put it another way, a Pareto efficient allocation is one for which each agent is as well off as possible, given the utilities of the other agents.
3. Suppose there is a fixed amount of two goods, $(\bar{x}, \bar{y})$, and there are only two individuals. In this case, a Pareto efficient allocation is one that maximizes the utility of agent 1, while holding agent 2 fixed at some given level of utility $\bar{u}$.

$$\max_{\{x_1, y_1\}} u_1(x_1) + y_1$$

$$s.t. u_2(\bar{x} - x_1) + \bar{y} - y_1 = \bar{u}.$$

4. Substituting from the constraint into the objective function, we have the unconstrained maximization problem

$$\max_{\{x_1\}} u_1(x_1) + u_2(\bar{x} - x_1) + \bar{y} - \bar{u}$$

which has the first-order condition:

$$u'_1(x_1) = u'_2(x_2). \tag{1}$$

5. Finally, consider the relationship between (1) and the competitive equilibrium. At an equilibrium price p^* , each consumer adjusts his consumption of x so that

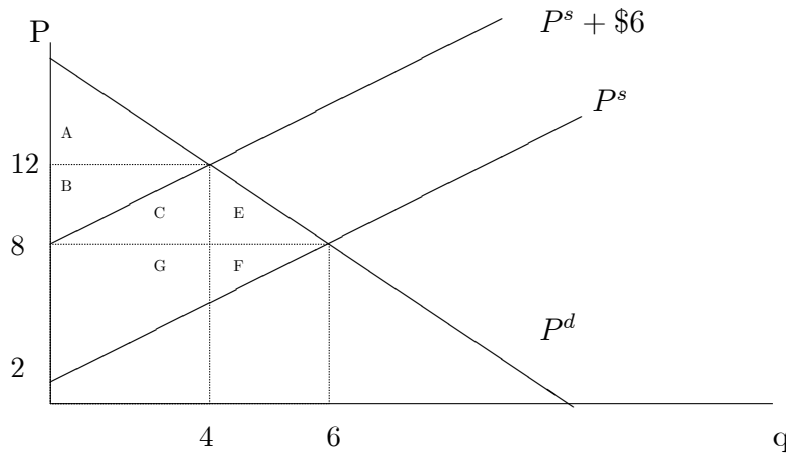
$$u'_1(x_1^*) = u'_2(x_2^*) = p^*.$$

Hence the necessary condition for Pareto efficiency is satisfied. Furthermore, any allocation that is Pareto efficient must satisfy (1), which essentially determines a price p^* at which this Pareto efficient allocation would be supported as a competitive equilibrium.

3 Competitive markets: Applications

1. The impact of an excise tax

- (a) An **excise tax** (or a specific tax) is an amount paid by either the consumer or the producer per unit of the good at the point of sale.
- (b) The effect of excise tax



- (c) deadweight loss with tax: E+F
- (d) Incidence of tax: The **incidence of tax** is the effect that the tax has on the prices consumers pay and sellers receive in a market. The amount by which the price paid by buyers, P^d , rises over the non-tax equilibrium price, P^* , is the incidence of the tax on consumers; the amount by which the price received by sellers, P^s , falls below P^* is called the incidence of the tax on producers.
- (e) The more inelastic side of the market bears the greater part of the tax burden.

$$\frac{\Delta P^d}{\Delta P^s} = \frac{\epsilon_{q^s, P}}{\epsilon_{q^d, P}}.$$

- (f) Example

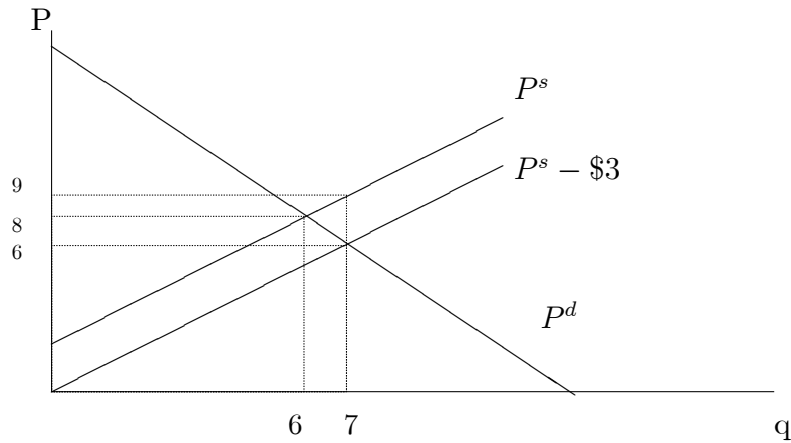
$$q^d = 10 - 0.5P^d,$$

$$q^s = \begin{cases} -2 + P^s, & P^s \geq 2 \\ 0, & P^s < 2. \end{cases}$$

With no tax, $P^* = 8, q = 6$. $CS = 36, PS = 18$.

With tax of \$6, $P^d = P^s + 6 = 12, P^s = 6$, and $q = 4$. $CS = 16, PS = 8$. Tax revenue=24. Deadweight loss=6

2. Subsidies: subsidies are *negative taxes*



Example:

$$q^d = 10 - 0.5P^d,$$

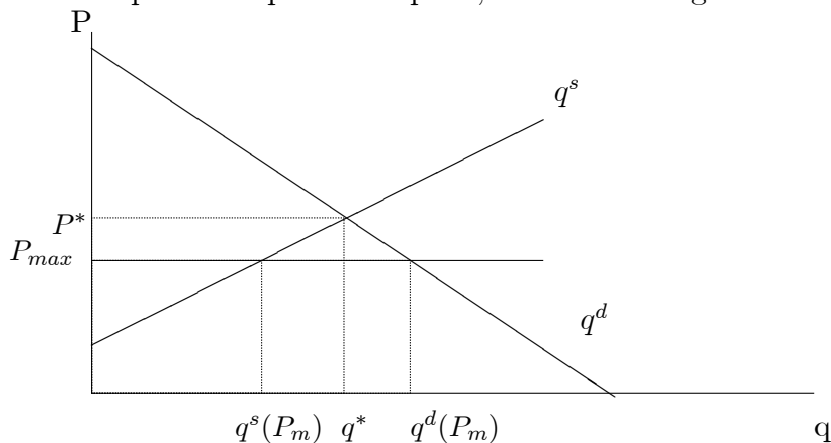
$$q^s = \begin{cases} -2 + P^s, & P^s \geq 2 \\ 0, & P^s < 2. \end{cases}$$

With no subsidy, $P^* = 8, q = 6$. $CS = 36, PS = 18$.

With subsidy of \$3, $P^d = P^s - 3, P^s = 9, P^d = 6$, and $q^d = 7$. $CS = 49, PS = 24.5$. Expenditure on subsidy=21. Deadweight loss=1.5.

3. Price ceiling

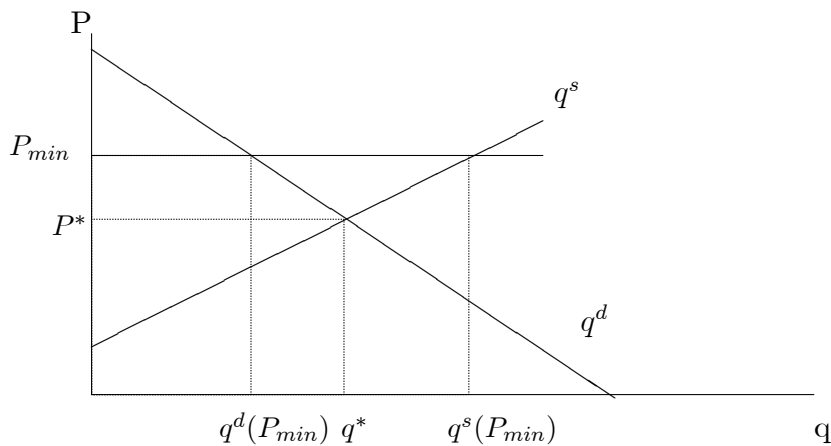
A **price ceiling** is a legal maximum on the price per unit that a producer can receive. If the price ceiling is below the pre-control competitive equilibrium price, then the ceiling is called **binding**.



- (a) Market not cleared;
- (b) Underproduction, and excess demand;
- (c) Deadweight loss.

5. Price floor

A **price floor** is a minimum price that consumers can legally pay for a good. Price floors sometimes are referred to as price supports. If the price floor is above the pre-control competitive equilibrium price, it is said to be **binding**.



- (a) Market not clear;
- (b) Excess supply and under-demand;
- (c) Deadweight loss.

4. Production Quotas: A **production quota** is a limit on either the number of producers in the market or on the amount that each producer can sell. The quota usually has a goal of placing a limit on the total quantity that producers can supply to the market.

